Robust Contactless Palmprint Verification with Deep Metric Learning: Checkpoint Report

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***Abstract*—Contactless palmprint verification offers a hygienic and user-friendly biometric modality, but real-world capture variations such as noise, motion blur, and lighting changes can severely degrade system performance. In this work, we develop a verification system using ResNet18 with ArcFace angular margin loss, trained and evaluated on the PolyU–IITD contactless palmprint dataset (611 subjects, 12,220 images) with subject-disjoint splits. Our baseline achieves an Equal Error Rate (EER) of 1.67% on clean test data. A systematic robustness benchmark across 9 corruption types at 5 severity levels reveals that noise (+1,170%), motion blur (+1,041%), and brightness variations (+796%) cause the most severe degradation, while contrast and scale changes remain manageable.**

I. Refined Problem Statement and Motivation

Contactless palmprint verification is an emerging biometric modality that captures palm images without physical contact, improving hygiene and user acceptance compared to fingerprint sensors. Palmprints contain rich discriminative features—principal lines, wrinkles, and texture patterns—that remain stable over time, making them well-suited for identity verification.

However, contactless capture introduces challenges absent in contact-based systems. Variations in hand pose, sensor distance, ambient lighting, motion blur, and compression artifacts all affect image quality. A practical verification system must not only achieve high accuracy on clean images but also degrade gracefully under these real-world conditions. When performance degrades, both security (increased false accepts) and usability (increased false rejects) are compromised.

This project addresses the question: *How robust is a deep learning-based palmprint verification system to common real-world capture variations, and which factors pose the greatest risk to deployment?*

A. Scope Changes from Proposal

The following changes have been made relative to the original proposal:

- **Dropped:** MobileNetV3 backbone comparison—we focused on ResNet18 to invest effort in a thorough robustness benchmark instead.
- **Completed:** Full robustness benchmark across 9 corruption types $\times$ 5 severity levels (45 conditions), exceeding the originally planned 4–6 types.
- **Deferred:** Mitigation implementation (robust augmentation training or quality gating) is deferred to the final phase of the project.

II. Data Engineering and Provisioning

A. Dataset

We use the **PolyU–IITD Contactless Palmprint Database v3.0** [1], comprising 611 subjects with 20 images each (10 left hand + 10 right hand), totaling 12,220 images. All images are 128×128 grayscale ROI-cropped palmprint regions.

B. Preprocessing

Each grayscale image is converted to 3-channel RGB for compatibility with ImageNet-pretrained backbones. Images are normalized using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). During training, we apply data augmentation including random cropping, horizontal flipping, rotation ($\pm15°$), color jitter, Gaussian blur, and random erasing to improve generalization.

C. Subject-Disjoint Splits

To prevent identity leakage, we use subject-disjoint splits where no subject appears in more than one partition. Each (subject, hand) combination is treated as a unique class, doubling the effective number of identities:

TABLE I
Dataset splits with subject-disjoint protocol.

Split	Subjects	Images	Classes
Train	427 (70%)	8,540	854
Validation	61 (10%)	1,220	122
Test	123 (20%)	2,460	246

A quality audit identified 4 empty placeholder subjects (H_ID608–611) in the original dataset; these were excluded from all splits.

III. Exploratory Data Analysis

A. Score Distributions

Analysis of the verification score distributions reveals strong separation between genuine and impostor pairs. Genuine pairs exhibit a mean cosine similarity of 0.759 ± 0.160, while impostor pairs have a mean of 0.033 ± 0.105. This bimodal separation, shown in Fig. 2, enables reliable threshold-based verification.

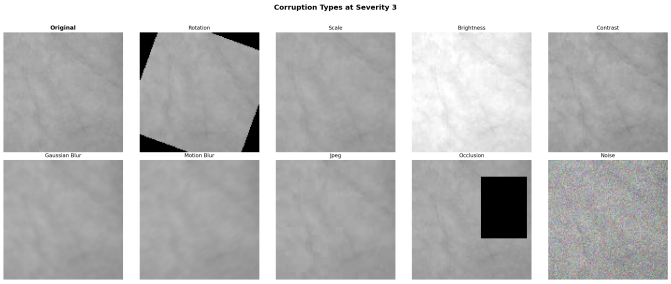

Fig. 1. Original palmprint image and 9 corruption types at severity level 3. Top row: Original, Rotation, Scale, Brightness, Contrast. Bottom row: Gaussian Blur, Motion Blur, JPEG, Occlusion, Noise.

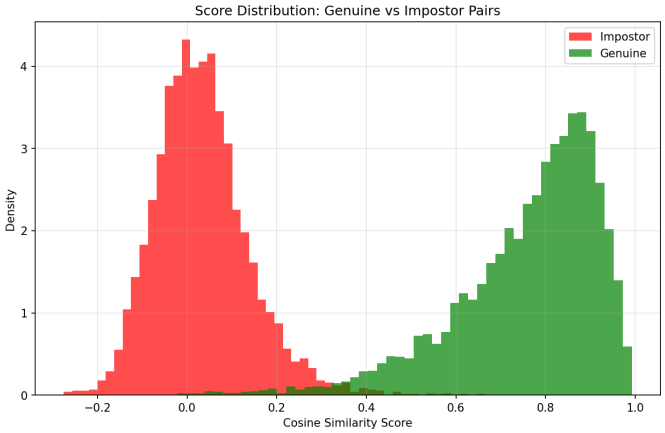

Fig. 2. Genuine (green) vs. impostor (red) cosine similarity score distributions on the test set. Clear separation enables reliable verification at a threshold of ∼0.30.

B. Embedding Space Visualization

Fig. 3 presents a t-SNE projection of the 256-dimensional embedding space for 20 randomly selected test identities. Each identity forms a tight, well-separated cluster, confirming that the ArcFace training objective successfully structures the embedding space for identity discrimination.

C. Class Balance

The dataset is perfectly balanced by design: each subject contributes exactly 10 images per hand, yielding uniform class sizes across all splits. This eliminates class imbalance as a confound in our evaluation.

IV. Baseline Implementation and Preliminary Results

A. Architecture

Our model uses a two-stage architecture with approximately 12M parameters:

1) **Backbone:** ResNet18 [3] pretrained on ImageNet, with the final fully-connected layer replaced by an identity mapping, yielding 512-D feature vectors.
2) **Projection Head:** A two-layer MLP ($512 \to 512 \to 256$) with batch normalization and ReLU, producing L2-normalized 256-D embeddings on the unit hypersphere.

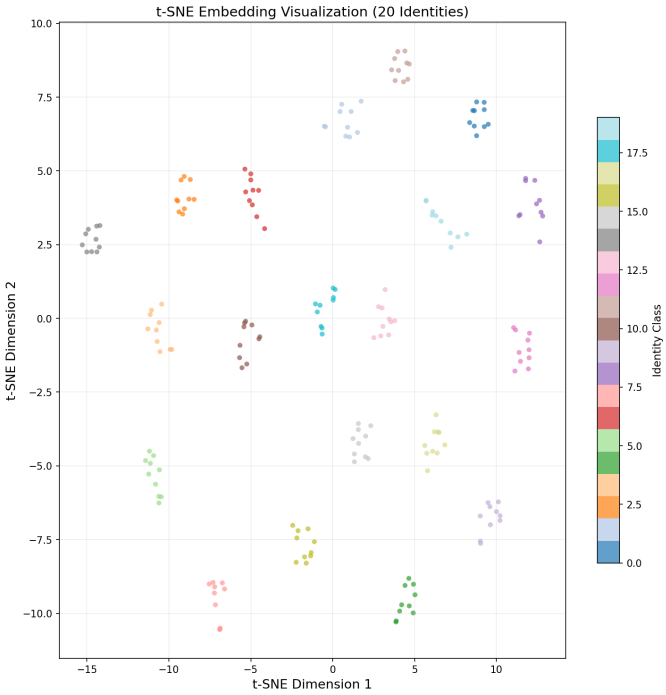

Fig. 3. t-SNE visualization of 256-D embeddings for 20 test identities. Tight, well-separated clusters confirm discriminative embedding learning.

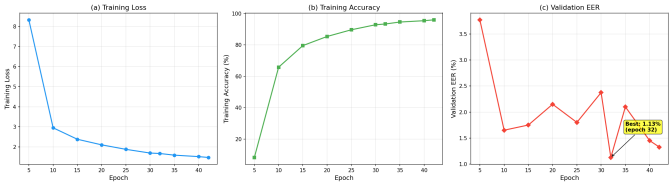

Fig. 4. Training convergence: (a) loss decreases from 8.3 to 1.5, (b) accuracy rises to 96%, (c) validation EER reaches minimum 1.13% at epoch 32.

During training, an ArcFace [2] classification head computes $\text{logit}_j = s \cdot \cos(\theta_j + m \cdot \mathbb{1}[j = y])$ with scale $s = 30$ and angular margin $m = 0.3$. The ArcFace head is discarded at inference; only the embedder is used for verification via cosine similarity.

B. Training

The model is trained with AdamW (lr = 3×10^{-4}, weight decay = 10^{-3}) and a cosine learning rate schedule for 50 epochs. Early stopping at epoch 42 selects the best checkpoint based on validation EER (1.13% at epoch 32). Fig. 4 shows the training convergence.

C. Clean Test Performance

Using the best checkpoint, we evaluate on the held-out test set (123 subjects, 246 classes, 2,460 images) with 5,000 genuine and 5,000 impostor pairs:

D. Threshold and Error Analysis

The EER operating point corresponds to a threshold of ∼0.30 on cosine similarity. The ROC curve (Fig. 5) confirms

TABLE II
VERIFICATION METRICS ON CLEAN TEST DATA.

Metric	Value
EER	1.67%
FAR @ 1% FRR	4.12%
FAR @ 0.1% FRR	48.88%
Genuine mean sim.	0.759 ± 0.160
Impostor mean sim.	0.033 ± 0.105

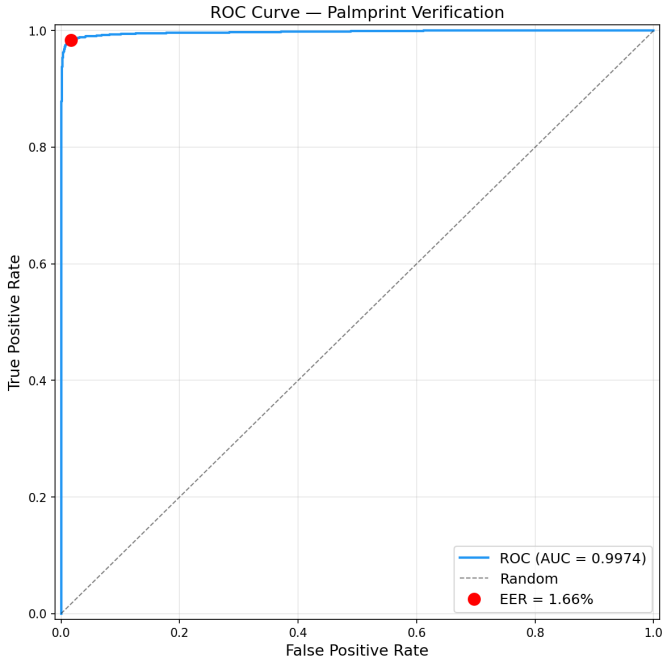

Fig. 5. ROC curve with EER operating point (red dot). AUC > 0.99 confirms strong baseline verification performance.

strong discriminative performance with AUC exceeding 0.99. However, the high FAR @ 0.1% FRR (48.88%) reveals that genuine and impostor distributions have overlapping tails—at very strict operating points, the system cannot reliably separate all pairs. This tail overlap likely stems from limited intra-class variability in the training data ($\sim$10 images per class).

E. Robustness Summary

Our robustness benchmark evaluates the model under 9 corruption types at 5 severity levels (45 conditions). Table III summarizes the results, ranked by worst-case degradation:

The three most damaging factors—noise, motion blur, and brightness—each cause $>5\times$ EER increase at maximum severity, presenting clear targets for mitigation in the final project phase.

V. CODE REPOSITORY

The complete source code is available at: `https://github.com/onggiahuy97/cs228-palmprint`.

Repository structure:

- `src/` – Training, evaluation, and corruption scripts

TABLE III
ROBUSTNESS BENCHMARK: EER DEGRADATION BY CORRUPTION TYPE.

Corruption	Mean EER	Worst EER	Degrad.
Noise	10.02%	21.90%	+1,170%
Motion Blur	8.46%	19.67%	+1,041%
Brightness	6.85%	15.45%	+796%
JPEG	5.12%	11.12%	+545%
Rotation	5.01%	9.15%	+430%
Occlusion	4.52%	7.80%	+352%
Gauss. Blur	2.85%	4.45%	+158%
Scale	2.41%	3.05%	+77%
Contrast	1.98%	2.28%	+32%

- `splits/` – Subject-disjoint split definitions (JSON)
- `figures/` – Generated plots and visualizations
- `checkpoints/` – Model weights and evaluation results
- `docs/` – Proposal and report documents
- `requirements.txt` – Python dependencies
- `README.md` – Setup and usage instructions

REFERENCES

[1] PolyU–IITD Contactless Palmprint Database v3.0, The Hong Kong Polytechnic University.

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[4] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. ICLR*, 2019.

[5] D. Zhong, X. Du, and K. Zhong, "Decade progress of palmprint recognition: A brief survey," *Neurocomputing*, vol. 328, pp. 16–28, 2019.

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